2015 MJC Prelim H2 Paper 2 Suggested Answers

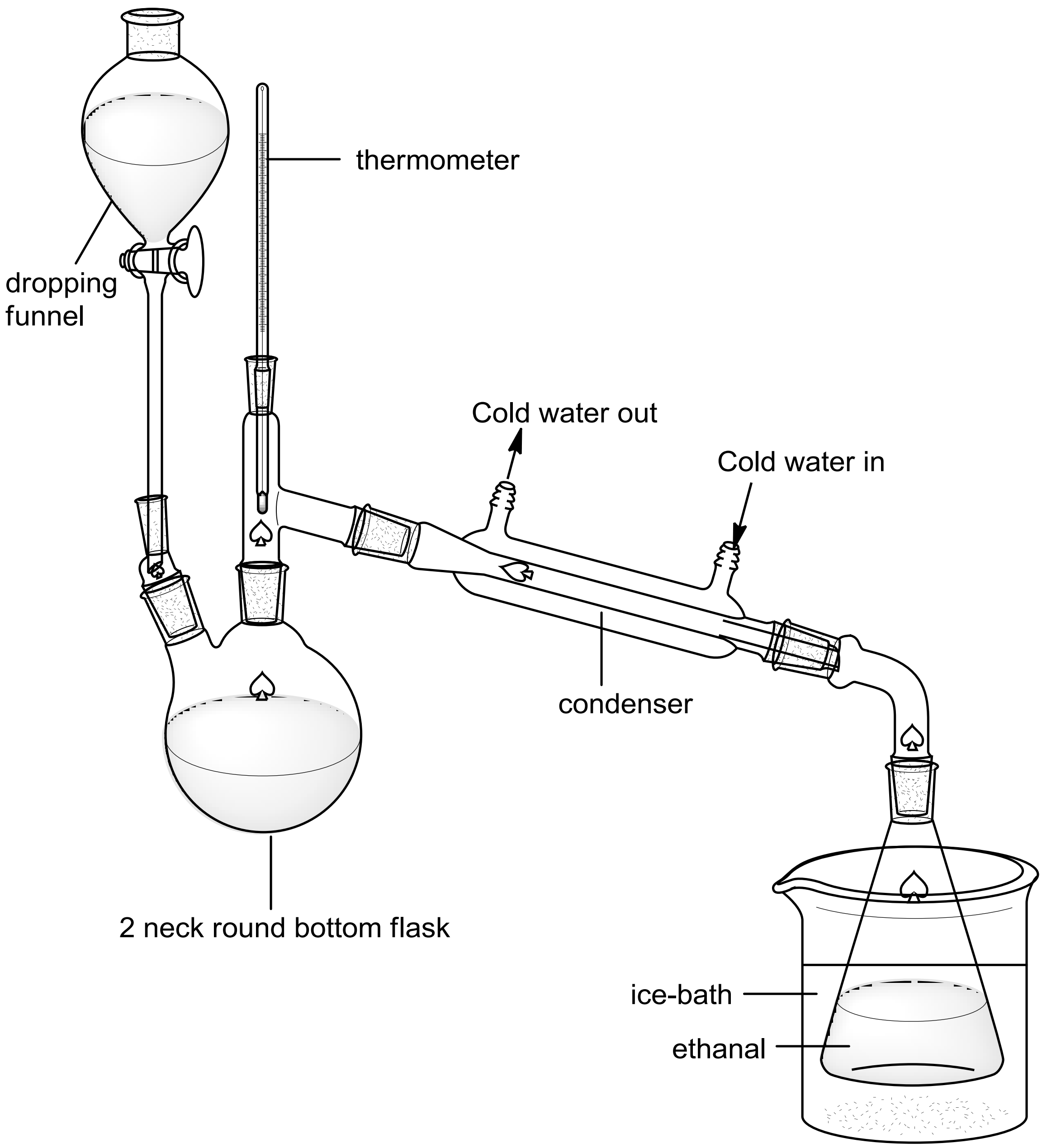
1(a) The reaction is very exothermic. The heat released during the reaction is sufficient to sustain the reaction.

(b)(i) $CH_3CH_2OH + [O] \rightarrow CH_3CHO + H_2O$

(ii) Number of mole of ethanol used = $\dfrac{3.945}{46.0}$ = 0.08576

Theoretical mass of ethanal produced = 0.08576 x 44.0 = 3.77 g

(c) 1) Set up the apparatus as shown below.



2) Using a 5 cm^3 measuring cylinder, measure 5 cm^3 of ethanol into a 50 cm^3 beaker.
3) Using a 10 cm^3 measuring cylinder, measure 6 cm^3 of water and pour it into the beaker.
4) Using a weighing balance, measure 5.0 g of potassium dichromate and dissolve into the water-ethanol mixture in the beaker.
5) Using a 10 cm^3 measuring cylinder, measure 9 cm^3 of dilute sulfuric acid and pour it into a 50 cm^3 round-bottom flask.
6) Weigh the mass of a 50 cm^3 conical flask to be used as a collection flask for ethanal.
7) Heat the sulfuric acid in the round-bottom flask using a heating mantle to about 60 ^oC. Then, switch off the heating mantle.
8) Pour the water-ethanol mixture containing dissolved potassium dichromate down the dropping funnel and add it dropwise into the sulfuric acid.
9) Use cold water to run through the condenser to condense the ethanal vapour formed.
10) Ethanal produced would be collected in the 50 cm^3 conical flask immersed in an ice-bath.
11) Reaction is completed when there is no more ethanal dripping from the condenser into the conical flask.
12) Reweigh the mass of the 50 cm^3 conical flask containing ethanal collected.

$$\text{Percentage yield} = \frac{\text{mass of ethanal collected}}{\text{theoretical mass of ethanal produced}} \times 100\%$$

(d) sodium hydroxide pellet

(e) organic compounds are flammable. Ensure no naked flame used.

2 (ai) Nuclear charge increases but screening effect remains fairly constant. Effective nuclear charge increases. Stronger electrostatic force of attraction between nucleus and valence electrons. Valence electrons are closer to the nucleus. Atomic radius decreases.

(ii) $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$

(iii) Aluminium oxide is an amphoteric oxide ; it undergoes neutralization with both acids and base to give a salt. Sulfur dioxide/trioxide is an acidic covalent oxide and undergoes neutralisation with alkali to give a salt and water.

(b)(i) No. of moles of monobasic H_2SO_3 = (0.24 – x)
No of moles of H^+ from monobasic H_2SO_3 = 0.24 – x
No of moles of H_2SO_4 = x
No of moles of H^+ from H_2SO_4 = 2x
No of moles of H^+ from a mixture of SO_2 and SO_3
= (0.24 – x) + 2x = 0.24 + x

(ii) No of moles of NaOH = $\frac{38.50}{1000} \times 1.00 = 0.0385$

No of moles of H^+ in 250 cm^3 = $\frac{250}{25.0} \times 0.0385 = 0.385$

(iii) 0.24 + x = 0.385 so x = 0.145

No of moles of SO_3 at equilibrium = 0.145 mol
No of moles of SO_2 at equilibrium = 0.095 mol
No of moles of O_2 at equilibrium = 0.0475 mol

$p_{SO_3} = \frac{0.145}{0.2875} \times 6 = 3.026$ atm

$p_{SO_2} = 1.983$ atm

$p_{O_2} = 0.9913$ atm

$$K_p = \frac{3.026^2}{1.983^2 \times 0.9913} = 2.35 \text{ atm}^{-1}$$

3 (ai) $\theta = +1.0 \div \frac{1}{37} \times \frac{2}{4} = +18.5°$

(ii) $\frac{1}{37} \div \frac{\text{q of Z}}{32} = \frac{1}{2.3}$

q of Z = 1.99 or 2
Z is negatively charged = $\underline{S^{2-}}$

(b) Atomic radius increases due to increasing number of quantum shells of electrons. Valence electrons are further away from the nucleus and more shielded by inner quantum shells of electrons. There is weaker electrostatic attraction between nucleus and the valence electrons. Hence, first ionisation energy generally decreases down the group.

(ci) V = Br_2
W = I_2

(ii) $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

(iii) Reducing agent.

(d) Down the group, H–X bond strength decreases. Bond energy decreases. Ease of breaking the H−X bond increases hence H_3O^+ and X^- can be formed more easily.

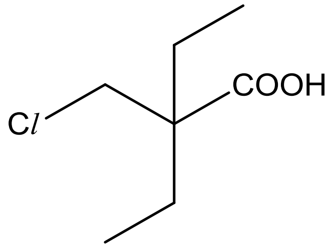
4 (ai) Step **I**: $KMnO_4$, dilute H_2SO_4, heat

Step **II**: KCN in ethanol, heat

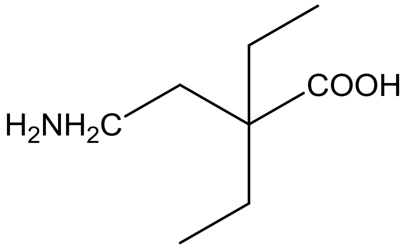
Step **IV**: anhydrous PCl_5, r.t.p

(ii)

K



M



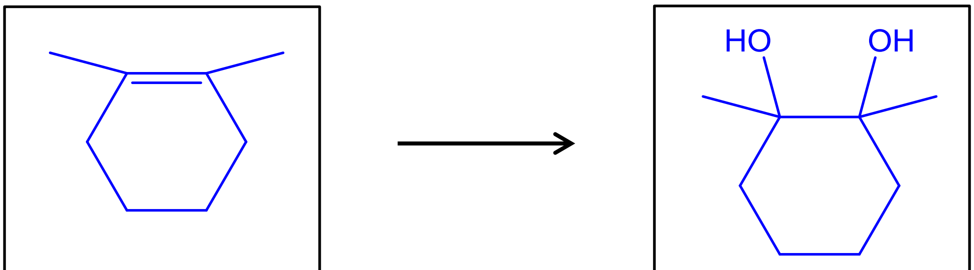
(bi) $K_{sp} = 3.01 \times 10^{-12} = 4s^3$

$s = 9.10 \times 10^{-5}$ mol dm^{-3}

(ii) $[K_2CrO_4] = 8.12 \times 10^{-3} / 0.500 = 0.01624$ mol dm^{-3}
Let s' be the new solubility of Ag_2CrO_4
$3.01 \times 10^{-12} = (2s')^2(s' + 0.01624)$. Assume $(s' + 0.01624) \approx 0.01624$
$3.01 \times 10^{-12} = (2s')^2(0.01624)$
$s' = 6.81 \times 10^{-6}$ mol dm^{-3}

(iii) Difference in Ag_2CrO_4 solubility = $9.10 \times 10^{-5} - 6.81 \times 10^{-6} = 8.42 \times 10^{-5}$ mol dm^{-3}
No of moles of Ag_2CrO_4 precipitated = $8.42 \times 10^{-5} \times 0.500 = 4.21 \times 10^{-5}$ mol
Mass of Ag_2CrO_4 precipitated = $8.138 \times 10^{-5} \times 332 = 0.0140$ g

5(ai)



compound **A** compound **B**

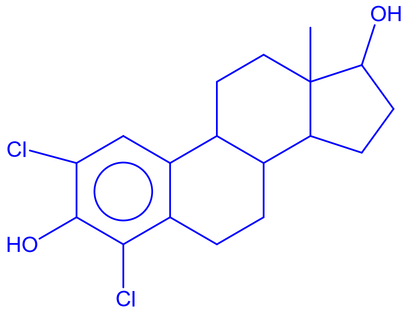
(aii) Step **I** : excess concentrated H_2SO_4, 180°C

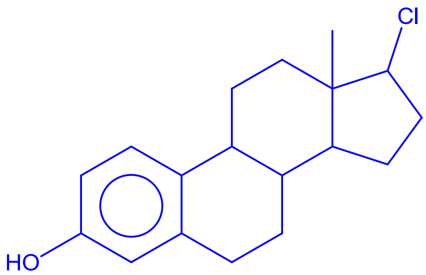
Step **II** : cold alkaline $KMnO_4$

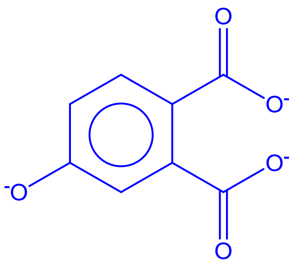
(b) Test: anhydrous PCl_5 , r.t.p.

Observations: White fumes of HC*l* for compound C.
No white fumes of HC*l* for androstenedione.

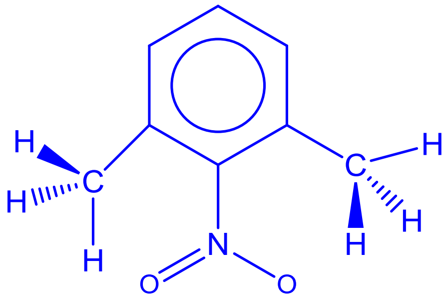
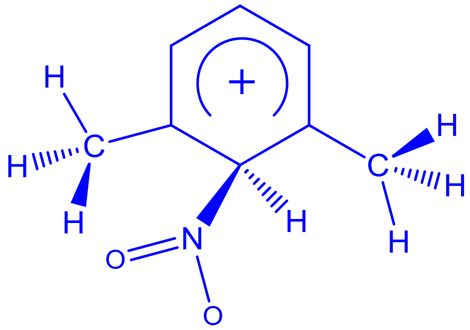
(c)(i) -(iii)





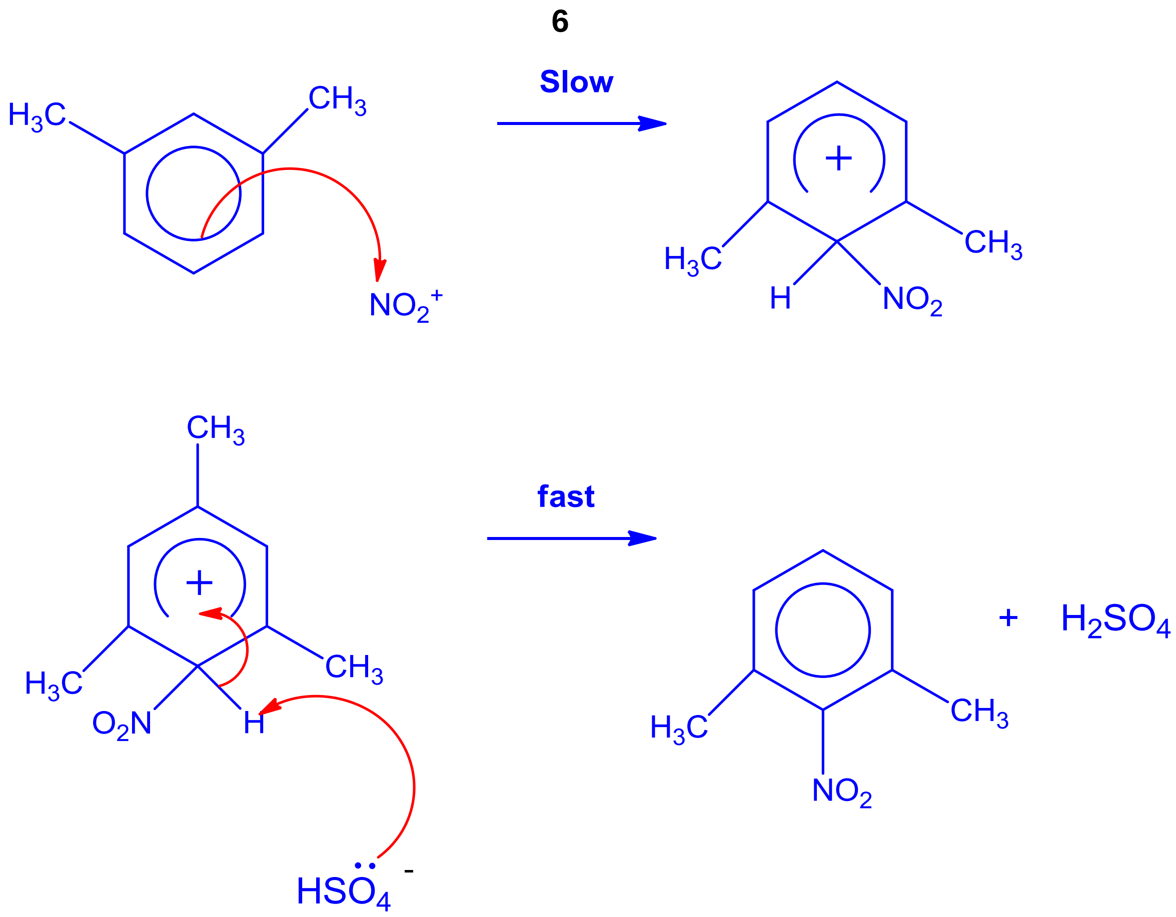


(di)

	
Isomer P	Intermediate R

(ii) Electrophilic substitution

$HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$

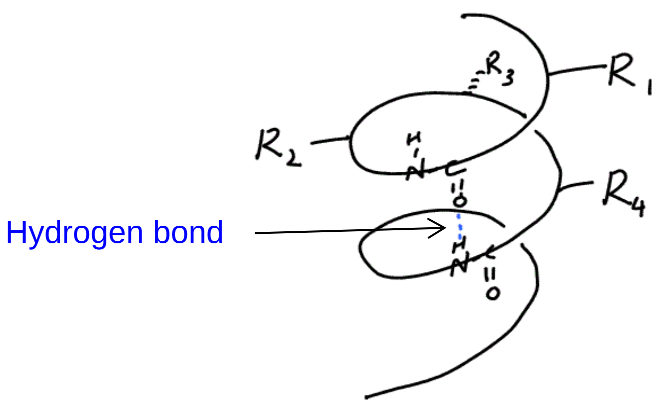


6(a) α-amino acid is a compound which has both the carboxylic acid group and the amine group attached to the same carbon atom.

(bi) lys or asp forms ion-dipole interactions with water while tyr or gln forms hydrogen bonding with water

(ii) Polar / hydrophilic R groups such as $–CH_2COOH$, $–(CH_2)_4NH_2$, $–CH_2CH_2CONH_2$ and $–CH_2C_6H_5OH$ (will be located on the outer surface of the globular protein. so that they could form hydrogen bonds or ion-dipole interactions with water. Non-polar / hydrophobic R groups such as $–CH_2CH(CH_3)_2$, $–CH_2SH$, groups will be located inside the protein away from the aqueous surroundings.

(iii)



The α-helix is a regular coiled configuration of the polypeptide chain, held in place by intra-chain hydrogen bonds.The O atom in amide group (C=O) of each amino acid is hydrogen-bonded to the H atom in amide group (N-H) of the fourth amino acid further down the chain. The R groups on the α-carbon point outside of the helix and are perpendicular to the main axis of the helix.

(c) Mr of X = 3(133) + 2(121.1) + 3(146) – 7(18) = 953.2

(d)(i) $–CH_2\text{-}S\text{-}H + H\text{-}S\text{-}CH_2– + [O] \rightarrow –CH_2\text{-}S\text{-}S\text{-}CH_2– + H_2O$

(ii) The covalent disulfide bonds are being destroyed; hence tertiary and quaternary structures will be affected.

(iii) Heating iron flatten hairs by providing energy to break the hydrogen bonds in keratin. Once water vapour from the atmosphere penetrates the hair fibre, it allows the hydrogen bonds to revert to their natural position.

(iv)